Supplementary Material

Techno-economic assessment of small modular reactors for artificial intelligence data centres and Bitcoin mining: dedicated versus shared supply with conventional and advanced reactor technologies

This document accompanies the article and follows its order. Section S1 completes the formulation of the model that Section 3 of the article summarises. Section S2 then lists the parameters and probability distributions, and Section S3 describes the electricity price data. With the inputs documented, Section S4 explains how the mining model was calibrated and Section S5 how the implementation was verified. Section S6 gathers the additional results to which the article refers. Finally, Section S7 describes the files with which every number can be reproduced. Monetary values are in constant 2025 USD.

S1. Model formulation

S1.1. Sets, time and notation

The formulation starts with the quantities that vary from hour to hour: the available capacity, the outages that determine it and the AI load. Hours are indexed by t within a simulated hourly record and project years by $y = 1, \dots, T$ after the start of operation. The plant has N modules of net capacity c; module i is available ($a_{i,t} = 1$) or not ($a_{i,t} = 0$), and the available capacity is

$$P_t^{av} = c \sum_{i=1}^{N} a_{i,t}. \tag{S1}$$

Availability is reduced by two kinds of outage. Planned outages of duration D^{pl} occur once per module every 24 months, so that $N/2$ modules are refuelled or maintained each year; they are scheduled back to back in two windows starting on day 80 and day 285. Forced outages, in turn, follow an alternating renewal process with exponentially distributed times to failure and to repair, with mean time to repair MTTR = 120 h and mean time to failure $\mathrm{MTTF} = \mathrm{MTTR}\,(1 - q)/q$, where q is the forced outage rate.

As for the AI load, it is $L_t = P^{AI} \ell_t$ with

$$\ell_t = \mathrm{clip}\left(\bar{\ell} + \delta_d \sin \frac{2\pi(h_t - 9)}{24} - \delta_s \cos \frac{2\pi(n_t - 15)}{365} + u_t,\ 0.30,\ 1 \right), \tag{S2}$$

where h_t is the hour of day, n_t the day of year, $\delta_d = 0.05$, $\delta_s = 0.03$, and $u_t = \varphi u_{t-1} + \sigma_u \sqrt{1 - \varphi^2}\, \xi_t$ is a first-order autoregressive disturbance with $\sigma_u = 0.05$, $\varphi = 0.97$ and ξ_t standard normal. In other words, the

load fraction is the sum of a daily cycle, a seasonal cycle and a persistent disturbance, limited to the interval between 0.30 and 1. The constant $\bar{\ell}$ is adjusted iteratively so that the mean of ℓ_t equals the target load factor.

S1.2. Hourly dispatch as a linear programme

The dispatch problem of Section 3.5 of the article is stated here in full, with one addition: a tier-1 firming resource, which represents the gas engines or the curtailed AI load of Section 3.6 of the article. With decision variables G_t (nuclear generation), m_t (mining), x_t (export), g_t (import), b_t^1 (tier-1 firming resource: gas engines or curtailed AI load) and b_t (emergency generation), the operator solves

$$\max \sum_t \left[\nu_t m_t + (p_t - \tau_x)x_t - c^{var}G_t - (p_t + \tau_g)g_t - c_1 b_t^1 - c_b b_t\right] \tag{S3}$$

subject to the energy balance

$$G_t + g_t + b_t^1 + b_t = L_t + m_t + x_t, \tag{S4}$$

to the capacity limits of generation, mining and export,

$$0 \le G_t \le P_t^{av}, \tag{S5}$$

$$0 \le m_t \le \tilde{M}, \tag{S6}$$

$$0 \le x_t \le \bar{X}, \tag{S7}$$

to the limits of the tier-1 resource and of imports,

$$0 \le b_t^1 \le \bar{B}_1, \tag{S8}$$

$$0 \le g_t \le \bar{G}, \tag{S9}$$

and to two conditions on the resources that cover a shortfall: emergency generation cannot be negative, and the three resources together cannot exceed the shortfall,

$$b_t \ge 0, \tag{S10}$$

$$g_t + b_t^1 + b_t \le s_t, \tag{S11}$$

with $s_t = \max(L_t - P_t^{av}, 0)$ and $\widetilde{M} = 0.97\,M$. Two remarks help to read these constraints. First, constraint (S11) reserves imported and backup energy for the shortfall of the AI load. This means that neither mining nor export uses non-nuclear energy. Second, when the tier-1 resource is curtailment of flexible AI load, b_t^1 is energy not delivered and is subtracted from the AI energy in the denominator of the LCAI.

Because the constraints do not couple hours, the problem separates by hour. Consequently, the optimum is a merit order, which is written here step by step. One condition on the firming resources is required: the tier-1 resource must cost no more than emergency generation, $c_1 \le c_b$, and, in any hour in which it is available together with grid imports, no more than imports, $c_1 \le p_t + \tau_g$. In the configurations studied the second part never binds, because the tier-1 resource appears only in islanded designs ($\bar{G} = 0$). The shortfall is covered first with the tier-1 resource,

$$b_t^1 = \min(s_t, \bar{B}_1), \tag{S12}$$

then with imports, when they cost no more than emergency generation,

$$g_t = \min(s_t - b_t^1, \bar{G})\,\mathbb{1}\left[p_t + \tau_g \le c_b\right], \tag{S13}$$

and finally with emergency generation,

$$b_t = s_t - b_t^1 - g_t. \tag{S14}$$

The surplus of nuclear capacity over the AI load served by the plant is

$$S_t = P_t^{av} - (L_t - s_t). \tag{S15}$$

This surplus is sent to the outlet that pays more and then to the other, provided that each of them pays more than the avoidable generation cost. If mining pays at least as much as export, $\nu_t \ge p_t - \tau_x$, mining is served first,

$$m_t = \min\left(S_t, \widetilde{M}\right)\mathbb{1}[\nu_t > c^{var}], \tag{S16}$$

and export takes what remains,

$$x_t = \min(S_t - m_t, \bar{X})\,\mathbb{1}[p_t - \tau_x > c^{var}]. \tag{S17}$$

Otherwise, export is served first,

$$x_t = \min(S_t, \bar{X})\, \mathbb{1}[p_t - \tau_x > c^{var}], \tag{S18}$$

and mining takes what remains,

$$m_t = \min\left(S_t - x_t, \tilde{M}\right) \mathbb{1}[\nu_t > c^{var}]. \tag{S19}$$

In both cases, nuclear generation follows from the energy balance (S4),

$$G_t = L_t - s_t + m_t + x_t. \tag{S20}$$

Two special cases complete the formulation. In the flexible-AI design the same order of priority is imposed for any idle-compute cost, including values above c_b, because the emergency generators are reserved for shortfalls that exceed the flexible load. As for the avoidable generation cost, it is $c^{var} = c^{vom} + \kappa_f c^{fuel}$, where $\kappa_f \in [0,1]$ is the share of fuel cost avoided when output is reduced ($\kappa_f = 1$ in the base case); the unavoidable part $(1 - \kappa_f)c^{fuel}$ is charged on available energy.

S1.3. Mining revenue and investment

With regard to mining, the annual frontier netback follows Eqs. (2)–(4) of the article, and what remains to be specified is when machines are bought and what the fleet earns. Let $v(y)$ be the year in which the fleet operating in year y was bought. At the start of each year the operator buys a new vintage if two conditions hold: the fleet is at least L years old (or absent), and the frontier netback observed at that date is at least R^{min}. Otherwise the existing fleet continues. The fleet netback is then

$$\nu_{y,t} = R_y^F \chi\, e^{-g\left(y - \frac{1}{2} - v(y)\right)} \frac{1 - f}{PUE_m}\, h_t, \tag{S21}$$

where h_t is constant within a day and follows $h_d = \exp(\sigma_h z_d)/\overline{\exp(\sigma_h z)}$ with z_d a first-order autoregressive process with daily autocorrelation 0.98 and $\sigma_h = 0.25$. On the cost side, the capital cost of the vintage bought in year v is $M\, k^A \left(R_v^F/\bar{R}\right)^\theta$; site infrastructure Mk^I is bought at the start of operation and refurbished at half cost every 20 years.

One computational device makes it possible to evaluate many netback paths quickly. The annual operating flows of each configuration (energies, revenues and costs from S1.2) are computed once on a grid of 73 values of the annual mean fleet netback (zero and 72 geometrically spaced values between 4 and 3,000

USD/MWh; 41 values up to 2,500 USD/MWh in the Monte Carlo analysis). They are then interpolated linearly for each year of each path; mining revenue is interpolated as revenue divided by the netback level.

S1.4. Life-cycle accounting

The annual flows are then converted into present values. With $d_y = (1+r)^{-(y-1/2)}$, a construction time of n months and uniform spending, the factor that compounds overnight cost to the start of operation is

$$\phi = \frac{1}{n}\sum_{j=0}^{n-1}(1+r)^{\left(n-j-\frac{1}{2}\right)/12}. \tag{S22}$$

On this basis, the present values at the start of operation are, for the nuclear plant,

$$PV(C^{nuc}) = (1-\iota)\,\phi\,I + \sum_y d_y\left(F + c^{var}G_y + (1-\kappa_f)c^{fuel}E_y^{av}\right) + (1+r)^{-T}D, \tag{S23}$$

for the mine,

$$PV\left(C^{mine}\right) = \sum_{v\in\mathcal{V}}(1+r)^{-v}\,Mk^A\left(\frac{R_v^F}{\bar{R}}\right)^{\theta} + \psi\,Mk^I + \sum_y d_y\,Mf^M, \tag{S24}$$

and for the firming resources,

$$PV\left(C^{firm}\right) = k^{ic}\max(\bar{X},\bar{G}) + \sum_y d_y\left(s^{sb}G^c + C_y^{imp} + c_b B_y + c_1 B_y^1\right) + \bar{B}_1\left(\psi_B\,k^B + \sum_y d_y\,f^B\right), \tag{S25}$$

where ι is the investment tax credit (zero in the base case), I the overnight cost, F the fixed O&M, D the decommissioning provision, $\mathcal{V}$ the set of purchase years, f^M the fixed O&M of the mine, k^{ic} the interconnection cost, s^{sb} the standby charge on the contracted backup capacity G^c (one module in regime G), and k^B, f^B the capital and fixed O&M cost of gas engines. The factors $\psi = 1 + 1/2\sum_{k=20,40,\ldots<T}(1+r)^{-k}$ and $\psi_B = 1 + 1/2\,(1+r)^{-20}$ account for the refurbishment of mining infrastructure and gas engines at half of their initial cost after 20 years. From these present values, the LCOE, LCAI and NPV follow Eqs. (17)–(19) of the article.

The number of simulated records differs between analyses. In the Monte Carlo analysis each draw uses one 8,760-hour record and one netback path, and mining capacities are held at their base-case optima (50

MW in the gas-backed design). In all other analyses each price year is paired with five independent outage and load realisations and results are means over 500 netback paths.

S1.5. Gradual build-out of the AI load

Finally, the build-out analysis of Section 4.8 of the article relaxes the assumption that the AI load is complete from the first year. In this analysis the contracted AI peak in year y is $\rho_y P^{AI}$ with $\rho = (0.25, 0.50, 0.75, 1, \ldots)$ for the three-year case and $(0.2, 0.4, 0.6, 0.8, 0.9, 1, \ldots)$ for the five-year case. As for the mine, its first vintage has capacity $M_1 \geq M^*$, chosen to maximise expected NPV, and later vintages have the steady-state optimum M^*; infrastructure is sized for M_1.

S2. Parameters and probability distributions

The parameters of the two reactors are given in Table 2 of the article. This section completes them. Table S1 lists the parameters of the mine, of the AI load and of the market, and Table S2 the probability distributions used in the Monte Carlo analysis. A note on the origin of the nuclear values follows the two tables.

Table S1. Mining, AI-load and market parameters. Ranges are the bounds of triangular distributions with the base value as mode.

Parameter	Unit	Base	Range	Basis
AI load factor	–	0.80	0.65–0.92	[1,2]
Diurnal / seasonal amplitude of AI load	–	0.05 / 0.03	fixed	assumed
Reference AI price for NPV	USD/MWh	100	fixed	[3]
Long-run median frontier netback, $\bar{R}$	USD/MWh	150	90–220	this study (Fig. 2)
Log standard deviation, σ; autocorrelation, ρ	–	0.40; 0.5	fixed	this study
Efficiency-progress rate, g	yr^{-1}	0.20	0.10–0.30	this study
Machine class relative to frontier, χ	–	0.95	fixed	[4]
Machine cost, k^A	million USD/MW	1.30	1.0–2.0	[4]
Elasticity of machine price to netback, θ	–	0.5	fixed	assumed
Mining site infrastructure, k^I	million USD/MW	0.70	0.4–1.0	[5]
Mining fixed O&M	USD/kW-yr	25	fixed	assumed
Pool fee; auxiliary overhead; uptime	–	1.5 %; 1.04; 97 %	fixed	assumed
Replacement interval, L; reinvestment threshold	yr; USD/MWh	5; 100	fixed	Table S5
Within-year log standard deviation of hashprice	–	0.25	fixed	[6]

Parameter	Unit	Base	Range	Basis
Gas backup: capital; fixed O&M; variable cost	USD/kW; USD/kW-yr; USD/MWh	1,800; 20; 50	fixed	assumed
Wholesale price level (scale factor)	–	1.0	0.7–1.6	assumed

Table S2. Input distributions of the Monte Carlo analysis (triangular: minimum, mode, maximum). Technology-specific inputs are drawn independently for the two reactors; common inputs are shared within a draw.

Input	iPWR	FHR
Overnight capital cost (USD/kWe)	lognormal, median 8,800, $\sigma = 0.25$	lognormal, median 8,800, $\sigma = 0.40$
Fixed O&M (USD/kW-yr)	triangular (130, 150, 238)	triangular (130, 165, 290)
Variable O&M (USD/MWh)	triangular (2.4, 2.9, 3.1)	triangular (2.4, 2.9, 3.1)
Fuel (USD/MWh)	triangular (11.0, 12.1, 13.3)	triangular (13.0, 18.0, 30.0)
Construction time (months)	triangular (43, 55, 71)	triangular (43, 60, 84)
Forced outage rate (%)	triangular (1.5, 3, 6)	triangular (2, 4, 12)
Real WACC (%), common	triangular (5, 7, 10)	same draw
AI load factor, common	triangular (0.65, 0.80, 0.92)	same draw
ERCOT price year, common	uniform over 2022–2025	same draw
Price scale factor, common	triangular (0.7, 1.0, 1.6)	same draw
$\bar{R}$, g, k^A, k^I, common	triangular (Table S1)	same draw
Frontier netback path, common	Eqs. (2)–(4) of the article, one path per draw	same draw

With regard to the nuclear values, the central ones are those of the ‘moderate’ case of the INL meta-analysis for a 300-MWe SMR between first and Nth of a kind (overnight cost 8,000 USD/kWe, fixed O&M 136 USD/kW-yr, variable O&M 2.6 USD/MWh, fuel 11.0 USD/MWh, construction 55 months; 2022 USD), escalated by 1.101. The ‘advanced’ and ‘conservative’ cases (5,500 and 10,000 USD/kWe; 118 and 216 USD/kW-yr) give the bounds for the iPWR [7]. To see where these cases lie in the distribution used, note that the lognormal distribution of overnight cost of the iPWR places the advanced and conservative values at its 7th and 81st percentiles and has a 95th percentile of 13,300 USD/kWe. The overrun case of 15,000 USD/kWe examined in Section 4.6 of the article, in turn, lies at its 98th percentile, and at the 91st percentile of the wider distribution of the FHR. For comparison, the estimate for the first-of-a-kind light-water SMR project cancelled in 2023 was about 20,000 USD/kWe [8]. As for the FHR, the fuel cost range reflects the cost of high-assay TRISO fuel and the absence of a commercial supply chain; the wider ranges for fixed O&M, construction time and forced outage rate are assumptions that reflect its lower technology readiness.

S3. Electricity price data

The price data support Sections 3.8 and 4.5 of the article. All series are observed hourly prices (Table S3). They are described below by region, and the conversion to a common currency follows.

For the United States, hourly day-ahead prices at the level of each regional transmission organisation are from the COVID-EMDA+ data hub [9], which covers January 2017 to November 2021 for ERCOT, PJM, CAISO, MISO, NYISO, ISO-NE and SPP. The series are complete except for CAISO in 2017 (the first 18 days are absent and were set to the median of that year), CAISO in 2020 (48 hours) and MISO in 2021 (24 hours), which were filled by interpolation over six hours and with the preceding value thereafter. Years 2017–2020 are calendar years and 2021 is December 2020 to November 2021. ERCOT requires a separate remark, because its series is extended beyond the period of the data hub. For ERCOT the data-hub series coincides with the hub-average day-ahead settlement point price published by the market operator (mean absolute difference of 0.01 USD/MWh in 2020). For this reason, the operator's series [10] is used for the calendar years 2020–2025 and for the twelve months to October 2026, and the data-hub series for 2017–2019.

For Brazil, the data are hourly spot prices (preço de liquidação das diferenças, PLD) by submarket from January 2021, when hourly pricing was introduced, to October 2026 [11]. The Southeast/Centre-West and Northeast submarkets are used; prices in the South and North submarkets are nearly identical to these (correlations of 0.98). The lowest price of each year, which is the regulatory floor, was 49.77, 55.70, 69.04, 61.07 and 58.60 BRL/MWh in 2021–2025.

For Europe, the data are hourly day-ahead prices by country from January 2015 to October 2026 [12], with one country-level series per country as published by the source. Luxembourg, which shares a bidding zone with Germany, is omitted. A market-year is included if at least 99 % of its hours are available, which gives 322 market-years in 30 countries; gaps are filled as above. Time stamps were shifted from UTC to local standard time.

All prices were then brought to a common basis. Prices in euros and reais were converted to US dollars at the average exchange rate of the year (1.1096, 1.1072, 1.1301, 1.1817, 1.1194, 1.1410, 1.1830, 1.0534, 1.0817, 1.0820 and 1.1306 USD per euro for 2015–2025; 5.3958, 5.1605, 4.9946, 5.3872 and 5.5891 reais per dollar for 2021–2025) and escalated to 2025 USD with the US consumer price index for all urban consumers (annual averages of 237.02, 240.01, 245.12, 251.11, 255.66, 258.81, 270.97, 292.66, 304.70 and 313.69 for 2015–2024 and 322.2 for 2025, a year in which no index was published for October, which was interpolated) [13]. The label 2026 denotes November 2025 to October 2026, for which the corresponding averages are 1.1584 USD per euro, 5.1862 reais per dollar and a price index of 331.3. Leap years are truncated to 8,760 hours.

One property of the model explains why these series can be used year by year. Because outages and the AI load are nearly independent of prices, the results for surplus energy depend on the distribution of prices and only weakly on their hourly order. Table S3 summarises the coverage and the annual mean prices of each market, and Fig. S1 shows price duration curves for the reference market and for selected market-years.

Table S3. Hourly price series by market: coverage and annual mean prices (2025 USD/MWh). In total 374 market-years in 39 markets; statistics of every market-year are in the file results/market_stats.csv.

Market	Region	Years	Market-years	Mean price	Lowest annual mean	Highest annual mean	Negative hours (highest year)
ERCOT	United States	2017–2026	10	55	27	173	54
PJM	United States	2017–2021	5	38	26	48	0
CAISO	United States	2017–2021	5	47	40	58	165
MISO	United States	2017–2021	5	34	25	42	0
NYISO	United States	2017–2021	5	36	24	45	0
ISONE	United States	2017–2021	5	44	29	56	0
SPP	United States	2017–2021	5	37	22	75	583
Brazil SE/CO	Brazil	2021–2026	6	32	13	62	0
Brazil NE	Brazil	2021–2026	6	30	12	59	0
Austria	Europe	2015–2026	12	102	43	305	332
Belgium	Europe	2015–2026	12	101	45	284	519
Bulgaria	Europe	2017–2026	10	119	56	293	170
Croatia	Europe	2018–2026	9	130	54	316	257
Czechia	Europe	2015–2026	12	101	46	287	322
Denmark	Europe	2015–2026	12	87	35	250	338
Estonia	Europe	2015–2026	12	88	47	224	198
Finland	Europe	2015–2026	12	68	40	179	725
France	Europe	2015–2026	12	98	46	320	507
Germany	Europe	2015–2026	12	95	43	274	511
Greece	Europe	2015–2026	12	121	64	325	496
Hungary	Europe	2015–2026	12	115	53	315	281
Ireland	Europe	2019–2026	8	142	53	262	382
Italy	Europe	2015–2024	10	124	55	352	0
Latvia	Europe	2015–2026	12	97	48	263	184
Lithuania	Europe	2015–2026	12	97	48	267	186
Montenegro	Europe	2024–2025	2	119	109	128	0
Netherlands	Europe	2015–2026	12	99	46	281	584
North Macedonia	Europe	2024–2024	1	112	112	112	0

Market	Region	Years	Market-years	Mean price	Lowest annual mean	Highest annual mean	Negative hours (highest year)
Norway	Europe	2015–2026	12	64	13	161	198
Poland	Europe	2015–2026	12	99	53	193	310
Portugal	Europe	2015–2026	12	92	48	195	540
Romania	Europe	2015–2026	12	112	49	308	196
Serbia	Europe	2017–2026	10	126	55	317	42
Slovakia	Europe	2015–2026	12	106	47	307	299
Slovenia	Europe	2015–2025	11	110	53	318	268
Spain	Europe	2015–2026	12	92	48	194	682
Sweden	Europe	2015–2025	11	59	29	139	609
Switzerland	Europe	2015–2026	12	111	48	327	303
United Kingdom	Europe	2017–2026	10	123	56	279	168

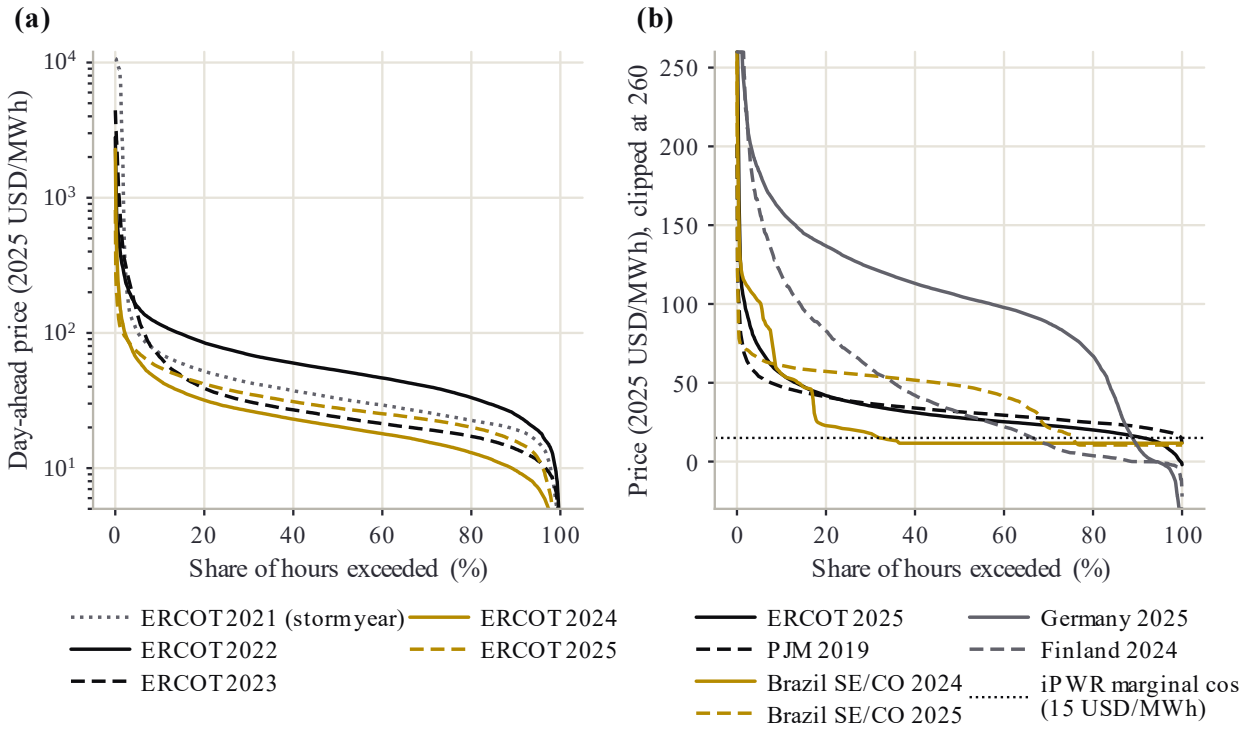


Fig. S1. Price duration curves. **(a)** ERCOT in the storm year 2021 and in the four years of the reference period. **(b)** Selected market-years.

S4. Calibration of the mining model

This section documents the three steps behind the mining model of Section 3.4 of the article: the data from which the frontier netback is computed, the zero-profit condition that the calibration satisfies, and the choice of machine class, replacement interval and reinvestment threshold.

The first step concerns the data. Daily miner revenue (block subsidy plus fees, in USD) and network hashrate from January 2016 to May 2026 are from Coin Metrics community data [6]. Hashprice is revenue divided by hashrate, expressed in USD per PH/s per day and smoothed with a 7-day mean. The frontier efficiency is that of the most efficient widely sold machine of the leading manufacturer at each date (manufacturer specifications; Table S4). Table S4 gives the resulting annual means.

Table S4. Annual mean hashprice and gross revenue per MWh of the frontier machine, with the machine generations that define the frontier efficiency.

Year	Mean hashprice (USD per PH/s per day)	Mean frontier netback (USD/MWh)	New frontier machine
2016	1,120	378	Antminer S9 (98 J/TH, Jun)
2017	1,170	498	–
2018	548	242	Antminer S15 (57 J/TH, Nov)
2019	218	217	Antminer S17 Pro (39.5 J/TH, Apr)
2020	114	145	Antminer S19 Pro (29.5 J/TH, May)
2021	315	445	–
2022	125	198	Antminer S19 XP (21.5 J/TH, Jul)
2023	75	145	–
2024	65	167	Antminer S21 (17.5 J/TH, Jan); Antminer S21 XP (13.5 J/TH, Sep)
2025	51	157	–
2026 (Jan–May)	35	153	Antminer S23 Hyd (9.5 J/TH, Jan)

The second step is the zero-profit condition, which gives the power price at which a stand-alone miner breaks even. Under the base parameters a machine of the value class earns 122 USD/MWh at the meter in its first year and 55 USD/MWh in its fifth (85 USD/MWh on average). The annuity of the machine cost over five years at a 7 % discount rate and 95 % utilisation is 38 USD/MWh. On this basis, a stand-alone miner paying a constant power price c has zero net present value over one machine life when $\sum_{y=1}^{5} d_y \left(\nu_y - c\right) h = k^A + \sum_y d_y \left(a^I + f^M\right)$, with h = 8,322 h/yr, a^I the annuity of infrastructure over 20 years and f^M the fixed O&M. This condition gives c^* = 40 USD/MWh, the value used as a reference in the article.

The third step is the choice of machine class, replacement interval and reinvestment threshold. Table S5 reports the mining credit of the islanded shared iPWR design for two machine classes (the current-generation value class at 1.30 million USD/MW and χ = 0.95, and a previous-generation machine at 0.50 million USD/MW and χ = 0.54, i.e. 17.5 J/TH at 8.8 USD per TH/s [4]), five replacement intervals and five reinvestment thresholds, with mining capacity fixed at 110 MW. Two results can be read from this table: the value class dominates, and the optimum is flat between four and six years. As for the reinvestment threshold, it has a small effect, whose sign depends on the process assumed. At 100 USD/MWh it lowers the expected credit by 0.6 USD/MWh under the base process, because the netback reverts to its mean and machines are cheap when it is low. Conversely, it raises the expected credit by 0.9 USD/MWh if the netback declines by 5 % per year. With a long-run netback of 110 USD/MWh, the expected credit is 0.8 USD/MWh without the threshold and 0.0 USD/MWh with it (file results/threshold_check.csv). For this reason, the threshold is retained as a conservative operating rule.

Table S5. Mean mining credit (USD per MWh of AI) of the islanded shared iPWR design by machine class, replacement interval and reinvestment threshold (USD/MWh of frontier netback).

Machine class	Replacement interval (yr)	Threshold 0	Threshold 80	Threshold 100	Threshold 120	Threshold 140
Current-generation value class	3	9.8	9.9	10.0	9.8	9.5
Current-generation value class	4	13.4	13.1	12.9	12.1	11.1
Current-generation value class	5	14.0	13.7	13.4	12.5	11.6
Current-generation value class	6	13.7	13.5	13.1	12.3	11.2
Current-generation value class	8	12.1	11.9	11.6	10.9	9.8
Previous generation	3	6.1	6.0	5.9	5.5	4.9
Previous generation	4	6.6	6.5	6.3	5.8	5.0
Previous generation	5	6.2	6.0	5.9	5.3	4.7
Previous generation	6	5.5	5.4	5.2	4.8	4.1
Previous generation	8	3.9	3.9	3.8	3.4	2.9

Fig. S2 completes the calibration: it compares the decay assumed in Eq. (S21) with the revenue realised by four machine generations.

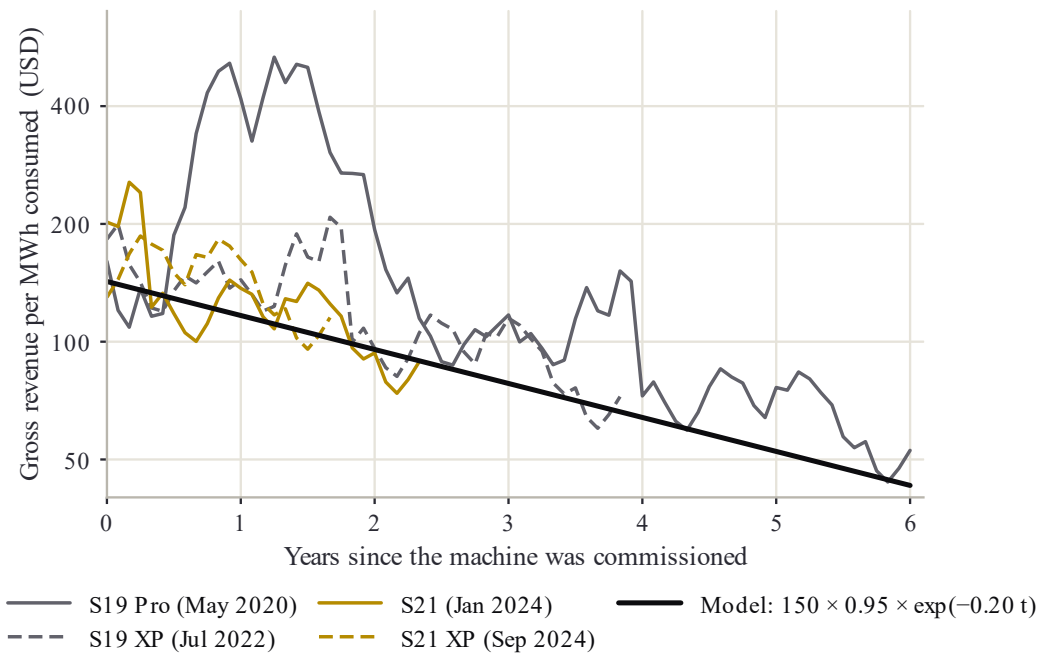


Fig. S2. Realised gross revenue per MWh of four machine generations from their commissioning date, compared with the decay assumed in the model for a median frontier netback.

S5. Verification

Three checks were made on the implementation: of the dispatch rule, of the simulated availability and of the sampling error. The dispatch rule was, in addition, proved formally, as Section S5.1 describes.

The first check concerns the dispatch. The closed-form dispatch (S12)–(S20) was compared with the solution of the linear programme (S3)–(S11) obtained with the HiGHS dual simplex solver (SciPy 1.18) for three 8,760-hour instances of the iPWR: a grid-connected campus (AI peak 300 MW, 80 MW of mining, export limit 120 MW, import limit 75 MW, CAISO 2019 prices, which include 165 negative hours; 43,800 variables); an islanded gas-backed campus (52,560 variables); and an islanded N−1 design in which the mining netback fluctuates around the avoidable generation cost. In the three instances, the objective values agree to a relative difference of 2×10^{-14} or less and the annual energies of all variables are identical (script code/verify_lp.py).

The second check concerns availability. Fig. S3 shows the distribution of available capacity in the 20-year base record: all four modules are available in 82 % (iPWR) and 81 % (FHR) of hours, three in 17 % and 18 %, and fewer than three in 1.6 % and 1.0 %. These last hours are the ones covered by emergency generation, which supplies 0.3 % (iPWR) and 0.2 % (FHR) of the AI energy in the islanded design.

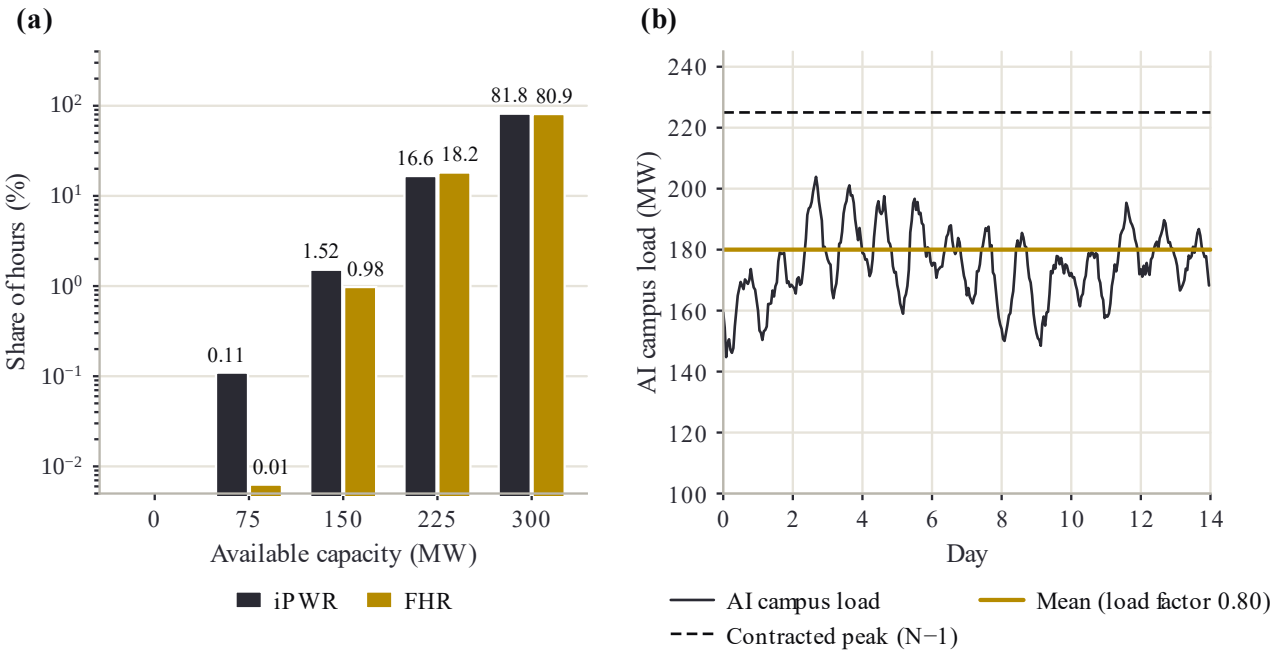


Fig. S3. **(a)** Distribution of available capacity in the base-case record. **(b)** Two weeks of the AI campus load in the islanded design.

The third check concerns sampling. Base-case results use 500 netback paths with common random numbers across configurations; the standard error of the mean mining credit of the islanded iPWR design is 0.3 USD/MWh.

S5.1. Formal verification of dispatch feasibility and optimality

The comparison with the linear programme covers three instances. To cover all of them, the feasibility and the optimality of the dispatch rule were also proved in the Lean 4 proof assistant [14] with the Mathlib library [15], so that every step is checked by the Lean kernel. The formalisation follows Section S1.2 line by line: the constraints (S4)–(S11), the hourly term of the objective (S3) and the rule (S12)–(S20), for data in any linearly ordered field and therefore for real numbers.

Two theorems are proved. Feasibility: the merit-order dispatch satisfies every constraint whenever the capacities and the AI load are non-negative. Optimality: under the condition on the firming resources stated in Section S1.2, no feasible dispatch earns a larger operating margin in the hour and, because no constraint links two hours, none earns a larger sum (S3) over any set of hours. Both cases of the proof rest on one elementary fact: when two uses with given unit values and capacities share a budget, serving the more valuable one first is optimal. Without shortfall the budget is the nuclear surplus, shared by mining and export; with shortfall it is the shortfall itself, shared by the firming resources.

The proof also made the hypotheses exact: the condition $c^{var} < c_1$ is not needed, whereas the condition on the tier-1 resource and imports cannot be dropped, as a counterexample in the same file shows. A further theorem states that in every feasible dispatch mining and export never coincide with imports, tier-1 firming or emergency generation, which is the precise sense in which the mine uses nuclear energy only. The proof concerns the rule as written in Section S1.2; that the code implements this rule is what the numerical comparison checks. The source (lean/SmrDispatch/Dispatch.lean) compiles with Lean 4.35.0 and Mathlib v4.35.0 without sorry and uses only the three standard axioms of Lean.

S6. Additional results

This section gathers the results to which the article refers, in the order in which the article uses them. It begins with the choice of mining capacity, which supports Section 4.3 of the article. In Fig. S4, the curve of each grid regime has a maximum, marked by a dot, which is the capacity adopted in the base case; beyond it, additional mining capacity lowers the result.

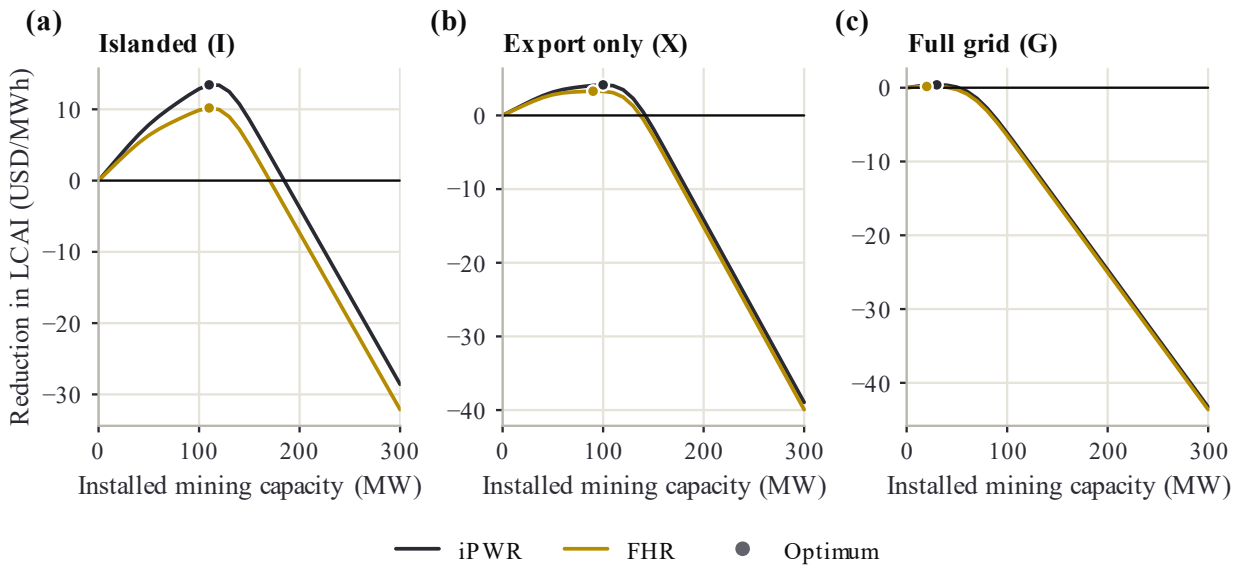


Fig. S4. Reduction in the break-even AI price as a function of installed mining capacity, by grid regime (base case, mean over 500 netback paths). Dots mark the optimum.

The comparison between mining and export of Section 4.5 of the article is detailed in two tables. Table S6 summarises it by market, and Table S7 gives the break-even AI price by market-year for the three markets that the article examines more closely: ERCOT, Brazil SE/CO and Germany.

Table S6. Market comparison by market: number of market-years in which mining is a better single outlet than export; mean value of an export connection, mean reduction in the break-even AI price when a mine is added in regimes X and G (USD/MWh), and number of market-years in which the optimal mine has zero capacity. The value of mining the surplus of the islanded design is 13.4 USD/MWh (iPWR) and 10.2 USD/MWh (FHR). Results for every market-year are in the files results/markets_iPWR.csv and results/markets_FHR.csv.

Market	Market-years	Mining better, iPWR	Mining better, FHR	Value of export, iPWR	Mine added to export, iPWR (X)	Mine added, iPWR (G)	Years with no mine, iPWR (X)
ERCOT	10	6	7	20.3	5.6	0.8	1
PJM	5	3	4	10.3	4.0	0.5	0
CAISO	5	1	2	15.9	1.7	0.1	1
MISO	5	5	5	7.9	5.3	0.7	0
NYISO	5	4	5	9.1	4.9	0.6	0
ISONE	5	3	3	13.7	2.9	0.3	0
SPP	5	4	4	11.0	6.8	1.0	0
Brazil SE/CO	6	5	5	8.3	6.8	1.2	1
Brazil NE	6	5	5	7.0	7.5	1.3	0
Austria	12	1	1	47.6	0.3	0.0	8
Belgium	12	0	1	46.8	0.1	0.0	10
Bulgaria	10	0	0	56.7	0.0	0.0	10
Croatia	9	0	0	63.4	0.0	0.0	8
Czechia	12	0	1	46.6	0.2	0.0	8

Market	Market-years	Mining better, iPWR	Mining better, FHR	Value of export, iPWR	Mine added to export, iPWR (X)	Mine added, iPWR (G)	Years with no mine, iPWR (X)
Denmark	12	3	4	39.1	1.0	0.1	8
Estonia	12	0	0	39.1	0.3	0.0	8
Finland	12	1	2	28.2	0.9	0.0	5
France	12	0	0	45.0	0.1	0.0	10
Germany	12	1	2	43.6	0.4	0.0	8
Greece	12	0	0	58.4	0.0	0.0	12
Hungary	12	0	0	54.5	0.0	0.0	11
Ireland	8	0	0	69.6	0.0	0.0	7
Italy	10	0	0	59.7	0.0	0.0	10
Latvia	12	0	0	43.9	0.1	0.0	9
Lithuania	12	0	0	44.3	0.1	0.0	9
Montenegro	2	0	0	57.1	0.0	0.0	2
Netherlands	12	0	1	46.2	0.2	0.0	10
North Macedonia	1	0	0	53.4	0.0	0.0	1
Norway	12	4	5	25.5	2.1	0.3	6
Poland	12	0	0	45.5	0.0	0.0	11
Portugal	12	0	0	42.2	0.1	0.0	11
Romania	12	0	0	52.9	0.1	0.0	11
Serbia	10	0	0	60.5	0.0	0.0	10
Slovakia	12	0	0	49.7	0.2	0.0	9
Slovenia	11	0	0	51.6	0.0	0.0	9
Spain	12	0	0	42.1	0.1	0.0	11
Sweden	11	3	5	23.0	1.7	0.2	4
Switzerland	12	0	0	52.3	0.1	0.0	11
United Kingdom	10	0	0	59.2	0.0	0.0	10

Table S7. Break-even AI price (USD/MWh) by market-year for regimes X and G in three markets, with the optimal mining capacity M (MW) for the iPWR. Islanded values do not depend on the market: A-I 187.4, C-I 174.0, D-I 198.0, F-I 187.8.

Market	Year	Mean price	A-X	C-X	M (C-X)	A-G	C-G	M (C-G)	D-X	F-X	D-G	F-G
ERCOT	2017	32.8	180.4	174.7	100	144.8	144.1	40	195.3	190.8	155.7	155.4
ERCOT	2018	42.4	175.3	170.9	100	143.4	143.0	40	190.4	187.1	154.4	154.3
ERCOT	2019	47.2	173.3	166.9	100	143.1	142.2	40	189.2	184.3	154.4	154.0
ERCOT	2020	27.1	183.7	174.8	100	145.9	144.3	40	197.4	190.7	156.3	155.5
ERCOT	2021	173.2	91.1	87.8	100	111.5	111.3	20	99.9	97.5	118.6	118.6
ERCOT	2022	70.8	159.1	159.1	0	138.5	138.5	0	175.1	175.1	150.2	150.2
ERCOT	2023	59.2	167.0	161.0	100	141.4	140.6	40	182.2	178.0	152.7	152.4
ERCOT	2024	28.9	181.3	172.9	100	144.8	143.4	40	194.9	188.9	155.2	154.6
ERCOT	2025	33.5	179.6	173.7	100	144.4	143.6	40	194.0	189.7	155.1	154.8
ERCOT	2026	33.6	180.3	173.4	100	144.9	143.8	40	194.6	189.6	155.6	155.2
Brazil SE/CO	2021	61.8	162.9	162.9	0	139.1	139.1	0	178.5	178.5	150.7	150.7
Brazil SE/CO	2022	12.6	189.2	176.1	120	147.2	144.7	40	200.8	192.0	156.8	155.5
Brazil SE/CO	2023	15.3	189.0	176.0	120	147.2	144.7	40	200.5	191.9	156.7	155.5
Brazil SE/CO	2024	24.5	183.6	174.3	100	145.8	144.2	40	195.8	190.0	155.5	154.9
Brazil SE/CO	2025	40.1	175.2	172.8	80	143.0	142.9	20	189.7	188.2	153.8	153.8
Brazil SE/CO	2026	39.9	175.0	172.2	80	142.7	142.6	20	189.3	187.5	153.6	153.6
Germany	2015	47.9	171.5	170.8	40	142.0	142.0	0	186.8	186.2	153.1	153.1
Germany	2016	43.0	174.2	172.4	60	142.8	142.8	0	189.3	187.9	153.8	153.8
Germany	2017	50.8	169.4	169.1	40	141.3	141.3	0	184.7	184.6	152.6	152.6
Germany	2018	65.9	161.2	161.2	0	138.8	138.8	0	177.1	177.1	150.4	150.4
Germany	2019	53.3	167.9	167.9	0	140.7	140.7	0	183.4	183.4	152.1	152.1
Germany	2020	43.1	173.2	171.6	60	142.3	142.3	0	188.4	187.3	153.6	153.6
Germany	2021	137.2	119.4	119.4	0	125.0	125.0	0	138.2	138.2	138.8	138.8
Germany	2022	274.3	41.0	41.0	0	100.4	100.4	0	64.4	64.4	116.7	116.7
Germany	2023	108.8	135.1	135.1	0	130.2	130.2	0	152.1	152.1	142.7	142.7
Germany	2024	86.5	147.8	147.8	0	134.2	134.2	0	164.2	164.2	146.3	146.3
Germany	2025	101.1	139.0	139.0	0	131.2	131.2	0	155.8	155.8	143.7	143.7
Germany	2026	125.6	125.8	125.8	0	127.5	127.5	0	143.1	143.1	140.3	140.3

With regard to sensitivity, Table S8 gives the one-way results behind Fig. 8 of the article, for both technologies.

Table S8. One-way sensitivities: break-even AI price (USD/MWh), re-optimised mining capacity M (MW, 20-MW grid) and mining credit (USD/MWh) for the iPWR (A, C) and the FHR (D, F). For fuel cost the two values are for the iPWR / FHR.

Parameter	Value	A-G	A-I	C-I	M	Credit	D-G	D-I	F-I	M	Credit
Base case	–	141.9	187.4	174.0	120	13.4	152.3	198.0	188.0	120	10.0
Overnight capital cost (USD/kWe)	6,050	108.4	142.7	129.3	120	13.4	118.3	152.6	142.6	120	10.0
Overnight capital cost (USD/kWe)	11,000	168.8	223.2	209.8	120	13.4	179.6	234.3	224.3	120	10.0
Real WACC (%)	5	115.3	152.3	137.5	120	14.8	124.9	161.9	150.5	120	11.3
Real WACC (%)	10	188.9	249.3	238.1	120	11.2	201.0	262.2	254.1	100	8.1
Fixed O&M (USD/kW-yr)	130	139.1	183.6	170.2	120	13.4	147.3	191.3	181.3	120	10.0
Fixed O&M (USD/kW-yr)	238	154.5	204.1	190.8	120	13.4	162.8	211.8	201.9	120	10.0
Fuel cost (USD/MWh)	9.1 / 13.5	138.4	184.4	169.4	120	15.0	147.3	193.5	181.1	120	12.4
Fuel cost (USD/MWh)	18.1 / 27.0	148.8	193.4	183.3	120	10.1	162.1	206.9	201.2	100	5.8
Construction time (months)	43	138.2	182.4	169.1	120	13.4	147.0	190.9	180.9	120	10.0
Construction time (months)	71	147.2	194.4	181.0	120	13.4	156.0	202.8	192.8	120	10.0
Forced outage rate (%)	1.5	141.5	186.8	171.9	120	14.9	151.8	197.7	185.5	120	12.2
Forced outage rate (%)	8.0	144.3	189.6	181.3	100	8.3	154.2	200.2	193.9	100	6.2
Economic life (years)	30	150.7	198.9	185.7	120	13.1	161.2	209.6	199.9	120	9.7
Economic life (years)	60	136.0	179.6	166.2	120	13.4	146.3	190.1	180.1	120	10.0
AI load factor	0.65	163.8	226.5	203.0	140	23.5	176.2	238.4	220.0	140	18.4
AI load factor	0.92	129.5	165.4	157.6	80	7.8	138.8	175.2	169.5	80	5.7
Long-run frontier netback (USD/MWh)	90	141.9	187.4	187.4	0	0.0	152.3	198.0	198.0	0	0.0
Long-run frontier netback (USD/MWh)	220	141.9	187.4	149.1	120	38.2	152.3	198.0	163.4	120	34.6
Efficiency-progress rate g (per year)	0.10	141.9	187.4	162.0	120	25.4	152.3	198.0	176.1	120	21.9
Efficiency-progress rate g (per year)	0.30	141.9	187.4	182.2	100	5.1	152.3	198.0	195.5	80	2.4
Machine cost (million USD/MW)	1.0	141.9	187.4	168.6	120	18.8	152.3	198.0	182.6	120	15.4
Machine cost (million USD/MW)	2.0	141.9	187.4	184.7	60	2.7	152.3	198.0	196.9	40	1.1
Mining infrastructure (million USD/MW)	0.4	141.9	187.4	172.1	120	15.2	152.3	198.0	186.1	120	11.8
Mining infrastructure (million USD/MW)	1.0	141.9	187.4	175.9	120	11.5	152.3	198.0	189.6	100	8.3
Avoidable share of fuel cost	0	142.1	194.5	174.5	120	19.9	152.9	208.5	188.8	120	19.7
Avoidable share of fuel cost	1	141.9	187.4	174.0	120	13.4	152.3	198.0	188.0	120	10.0

Table S9, in turn, reports the structural stress tests discussed in Section 4.7 of the article.

Table S9. Structural stress tests: mining credit (USD per MWh of AI) of the islanded shared design with mining capacity fixed at its base-case optimum. The reinvestment rule of Section S1.3 applies, so that no machines are bought once revenue has collapsed.

Stress case	iPWR mean	iPWR 5th–95th pct.	FHR mean	FHR 5th–95th pct.
Base	13.4	3.9 to 24.3	10.2	0.9 to 20.9
Mining revenue falls to zero from year 3	−5.3	−10.2 to 0.5	−5.8	−10.5 to 0.0
Mining revenue falls to zero from year 8	0.3	−6.7 to 9.2	−1.0	−7.9 to 7.8
Mining revenue falls to zero from year 15	6.5	−2.0 to 16.9	4.3	−4.0 to 14.6
Frontier netback declines by 2 % per year	5.3	−2.6 to 14.9	2.4	−5.3 to 11.7
Frontier netback declines by 5 % per year	−0.1	−6.4 to 7.7	−2.2	−8.1 to 5.5
One three-year bull cycle (netback × 3) in years 6–8	25.3	11.5 to 41.6	22.0	8.2 to 38.0

Table S10 shows how the results of the export-only regime change with the export limit.

Table S10. Effect of the export limit on the islanded-firmness designs of the iPWR (ERCOT 2017–2020): break-even AI price (USD/MWh). The interconnection cost scales with the limit.

Export limit (MW)	A-X	C-X	Mining capacity (MW)	Energy used, A-X (%)	Energy used, C-X (%)
0	187.4	174.0	110	63	95
15	184.9	172.2	110	67	98
30	182.4	170.9	110	71	99
60	177.8	168.9	100	79	99
100	172.2	166.6	100	89	99
150	170.4	166.2	100	93	99
225	170.9	166.7	100	93	99
300	171.4	167.3	100	93	99

Tables S11 and S12 give the results behind Fig. 11 of the article: the number of modules and the gradual build-out of the AI load.

Table S11. Plant granularity: break-even AI price (USD/MWh) for plants of 2 to 12 modules of 75 MWe with an N−1 design in the islanded regime.

Technology	Modules	Plant (MWe)	AI-only, islanded	Shared, islanded	Mining (MW)	Mining credit	AI-only, grid-backed
iPWR	2	150	273.2	241.7	75	31.5	143.7
iPWR	3	225	209.0	190.5	101	18.5	142.8
iPWR	4	300	187.4	174.0	120	13.4	141.9
iPWR	6	450	169.8	161.4	135	8.4	141.6
iPWR	8	600	162.4	155.2	150	7.2	141.1
iPWR	12	900	155.6	149.6	180	6.0	140.6
FHR	2	150	286.9	262.9	75	24.0	154.6
FHR	3	225	220.4	206.8	101	13.6	153.2

Technology	Modules	Plant (MWe)	AI-only, islanded	Shared, islanded	Mining (MW)	Mining credit	AI-only, grid-backed
FHR	4	300	198.0	187.9	105	10.1	152.3
FHR	6	450	180.4	174.2	112	6.2	151.9
FHR	8	600	172.8	167.9	120	4.9	152.2
FHR	12	900	165.8	161.8	180	4.0	151.5

Table S12. Gradual build-out of the AI load: break-even AI price (USD/MWh) and capacity of the first and later mining vintages.

Technology	Regime	Build-out	AI only	Shared	Credit	First vintage (MW)	Later vintages (MW)
iPWR	I	immediate	187.4	174.0	13.4	110	110
iPWR	I	3-year	206.5	190.2	16.3	140	110
iPWR	I	5-year	214.5	196.6	17.8	140	110
iPWR	G	immediate	141.9	141.6	0.4	30	30
iPWR	G	3-year	152.4	151.6	0.8	30	30
iPWR	G	5-year	156.8	155.9	0.9	30	30
FHR	I	immediate	198.0	187.8	10.2	110	110
FHR	I	3-year	217.7	205.0	12.6	110	110
FHR	I	5-year	225.9	212.1	13.8	140	110
FHR	G	immediate	152.3	152.2	0.2	20	20
FHR	G	3-year	163.8	163.4	0.4	20	20
FHR	G	5-year	168.6	168.1	0.5	20	20

Finally, Table S13 and Fig. S5 give the Monte Carlo distributions for all configurations, which Section 4.7 of the article summarises.

Table S13. Monte Carlo results (5,000 draws): levelised cost of electricity and break-even AI price by configuration.

Indicator (USD/MWh)	iPWR mean	iPWR 5 %	iPWR median	iPWR 95 %	FHR mean	FHR 5 %	FHR median	FHR 95 %
LCOE	135	95	132	187	155	95	146	247
AI only, islanded (A/D-I)	207	141	201	294	230	135	216	376
Shared, islanded (C/F-I)	196	127	191	284	224	126	210	369
AI only, export only (A/D-X)	188	121	183	277	216	120	202	362
Shared, export only (C/F-X)	187	118	182	274	216	118	201	363
AI only, gas-backed	166	116	161	232	185	113	175	294
Shared, gas-backed	164	114	160	229	185	113	174	293
AI only, grid-backed (A/D-G)	156	107	152	221	176	106	166	286
Shared, grid-backed (C/F-G)	156	107	152	221	177	107	166	286

The same analysis covers the dedicated configurations of Section 4.2 of the article. For a reactor dedicated to mining, the net present value per kW has a median of −10,219 USD/kW (iPWR, islanded) and −11,412 USD/kW (FHR) and is positive in 0.4 % and 0.4 % of draws. For merchant sales, in turn, the medians are −9,301 and −10,282 USD/kW, positive in 0.5 % and 0.4 % of draws.

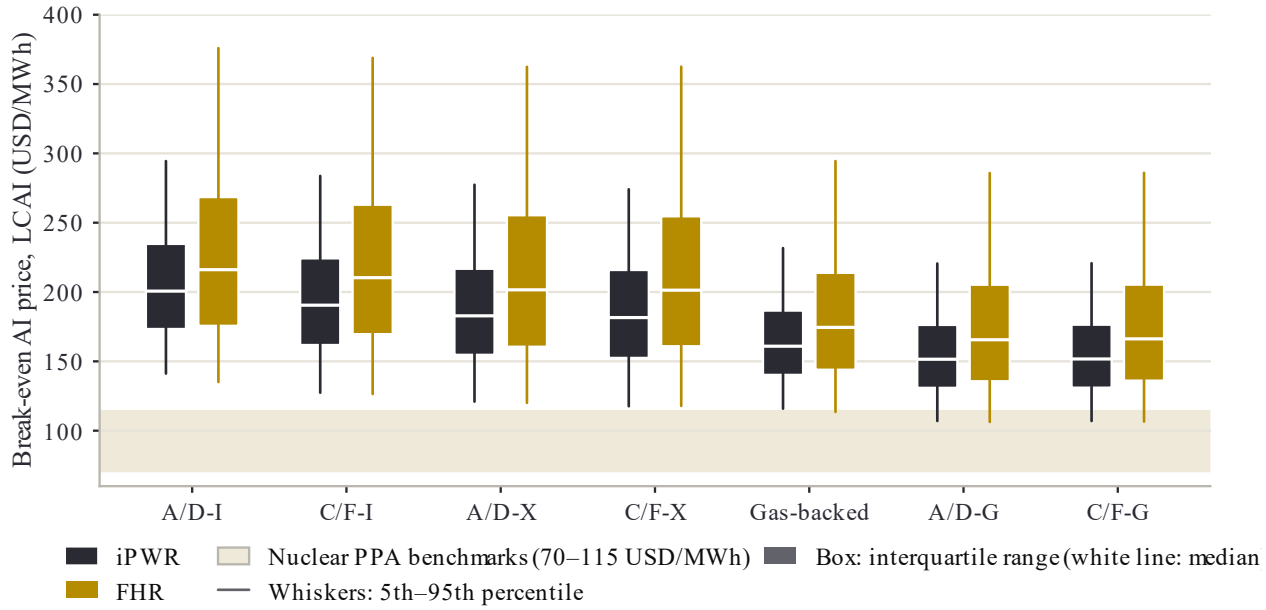


Fig. S5. Monte Carlo distributions of the break-even AI price by configuration: median (white line), interquartile range (box) and 5th–95th percentile range (whiskers).

S7. Reproducibility

The model source code, input datasets, numerical results and formal verification files are publicly available in the companion GitHub repository:

https://github.com/lucasfffaria/Technoeconomic_analysis_SMR_AI_Bitcoin

The repository's README.md provides a detailed description of the files, software versions, dependencies and commands required to reproduce the analyses. The repository is organised as follows:

- **code/model.py:** parameters, availability and load generators, hourly dispatch, mining netback process and life-cycle evaluation.
- **code/prepare_prices.py:** conversion of the files downloaded from the market operators into compact hourly tables.
- **code/markets.py:** loading of the price series, currency conversion and definition of the market-years.
- **code/common.py:** shared helpers (hourly records, netback paths, expectation over paths and optimisation of mining capacity).

- **code/run_base.py, run_firming.py, run_markets.py, run_sens.py and run_mc.py:** analyses underlying the numerical results presented in the article and supplementary material; outputs are written to results/ as CSV and JSON files.
- **code/verify_lp.py:** comparison of the closed-form dispatch with the linear programme (Section S5).
- **code/export_figure_data.py:** generation of the data series underlying the figures computed from the raw data or model outputs (hashprice and frontier netback, dispatch example, price duration curves, availability and AI load).
- **code/facts.py:** collection of the numerical values quoted in the article and supplementary material into results/facts.json.
- **lean/:** Lean 4 formalisation of the hourly dispatch (Section S5.1). SmrDispatch/Dispatch.lean contains the definitions and proofs, SmrDispatch/Audit.lean reports the axioms on which each theorem depends, and check_statements.py independently tests the statements against the linear programme on random instances.
- **data/raw/:** hourly electricity prices from the COVID-EMDA+ data hub, ERCOT, CCEE and Ember, and daily Bitcoin network data from Coin Metrics community data, retrieved in November 2026. The compact ERCOT, CCEE and Ember tables were generated using prepare_prices.py. The original downloads used to construct these tables are not redistributed but remain obtainable from their respective providers, subject to the applicable terms of use.
- **results/:** numerical results of the analyses, including the data underlying the tables and figures of the article and supplementary material.

Random seeds are fixed, and the README.md specifies the software versions used to generate the reported results. Repeating the analyses with the documented software environment reproduces the numerical outputs; minor numerical differences may arise with other library versions. The comparison across market-years is the most computationally intensive step.

To evaluate other hourly electricity price series, a loader can be added to markets.py that returns an array of 8,760 hourly prices expressed in constant 2025 USD/MWh. The market analysis can then be repeated by running run_markets.py separately for the iPWR and FHR technologies.

References

[1] I. Latif, A.C. Newkirk, M.R. Carbone, A. Munir, Y. Lin, J. Koomey, X. Yu, Z. Dong, Single-node power demand during AI training: Measurements on an 8-GPU NVIDIA H100 system, IEEE Access 13 (2025) 61740–61747. https://doi.org/10.1109/ACCESS.2025.3554728.

[2] F. Acun, C. Hankendi, E. Levine, H. Reynolds, J. Bardwick, A.K. Coskun, Investigating power consumption flexibility of AI data centers for demand response participation, in: Proceedings of the 17th ACM International Conference on Future and Sustainable Energy Systems, 2026: pp. 344–348. https://doi.org/10.1145/3744255.3798112.

[3] B. Martucci, Meta–Constellation PPA could be first of many deals for existing reactors, (2025). https://www.utilitydive.com/news/meta-constellation-ppa-could-be-first-of-many-deals-for-existing-reactors/750567/ (accessed November 10, 2026).

[4] MillionMiner, Bitcoin mining update November 2026: Revenue and difficulty, (2026). https://millionminer.com/news/bitcoin-mining-update-november-2026 (accessed November 10, 2026).

[5] Cointelegraph, Bitcoin miners, AI data centers and power infrastructure (reporting the CoinShares first-quarter 2026 mining report), (2026). https://cointelegraph.com/magazine/bitcoin-miners-ai-data-centers-power-infrastructure (accessed November 10, 2026).

[6] Coin Metrics, Community network data: Bitcoin daily metrics, (2026). https://github.com/coinmetrics/data (accessed November 10, 2026).

[7] A. Abou-Jaoude, L.M. Larsen, N. Guaita, I. Trivedi, F. Joseck, C. Lohse, E. Hoffman, N. Stauff, K. Shirvan, A. Stein, Meta-analysis of advanced nuclear reactor cost estimations. Report INL/RPT-24-77048, Rev. 0, Idaho National Laboratory, Idaho Falls, ID, 2024.

[8] Abou-Jaoude, A. et al., Literature review of advanced reactor cost estimates. Report INL/RPT-23-72972, Rev. 3, Idaho National Laboratory, Idaho Falls, ID, 2023.

[9] G. Ruan, D. Wu, X. Zheng, H. Zhong, C. Kang, M.A. Dahleh, S. Sivaranjani, L. Xie, A cross-domain approach to analyzing the short-run impact of COVID-19 on the US electricity sector, Joule 4 (2020) 2322–2337. https://doi.org/10.1016/j.joule.2020.08.017.

[10] Electric Reliability Council of Texas, Historical DAM load zone and hub prices (report NP4-180-ER), 2020–2026, (2026). https://www.ercot.com/mp/data-products/data-product-details?id=NP4-180-ER (accessed November 10, 2026).

[11] Câmara de Comercialização de Energia Elétrica (CCEE), Preço de liquidação das diferenças horário por submercado, 2021–2026. Portal Dados Abertos CCEE, (2026). https://dadosabertos.ccee.org.br/dataset/pld_horario (accessed November 10, 2026).

[12] Ember, European wholesale electricity price data (hourly day-ahead prices by country), (2026). https://ember-energy.org/data/european-wholesale-electricity-price-data/ (accessed November 10, 2026).

[13] Federal Reserve Bank of St. Louis, FRED economic data: Consumer price index for all urban consumers (CPIAUCNS) and foreign exchange rates (AEXUSEU, AEXBZUS, EXUSEU, EXBZUS), (2026). https://fred.stlouisfed.org (accessed November 10, 2026).

[14] L. de Moura, S. Ullrich, The Lean 4 theorem prover and programming language, in: Automated Deduction – CADE 28, Springer, Cham, 2021: pp. 625–635. https://doi.org/10.1007/978-3-030-79876-5_37.

[15] The mathlib Community, The Lean mathematical library, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs, 2020: pp. 367–381. https://doi.org/10.1145/3372885.3373824.